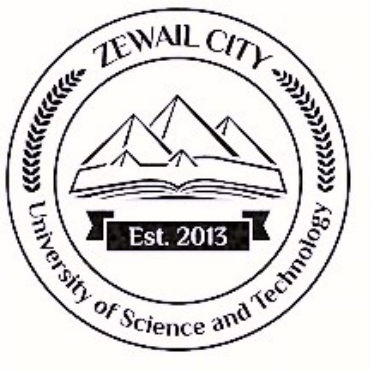
Tutorial 2: Exact and Non-Exact ODEs

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1 Important Points

Let us first review some important points that are essential for understanding of the topic. Linear first order ODEs of the form,

$$y'(x) + p(x)y(x) = q(x)$$

has a solution which can be written in the form,

$$y(x) = y_0 e^{-\int_{x_0}^{x} p(t)dt} + \int_{x_0}^{x} q(t) e^{-\int_{t}^{x} p(s)ds} \tag{1}$$

Suppose that we have an Exact DE with an integrating factor μ,

$$\mu M + \mu N y' = 0$$

Since this DE is exact,

$$(\mu M)_y = (\mu N)_x$$

Producing,

$$N\mu_x - M\mu_y = \mu(M_y - N_x)$$

Suppose that μ is a function of $v(x, y)$. Using chain rule we can transform the previous equation into,

$$\frac{1}{\mu}\frac{d\mu}{dv} = \frac{M_y - N_x}{v_x N - v_y M} \tag{2}$$

Thus, if the right hand side is only a function of v, $\phi(v)$. We can easily find the integration factor from,

$$\mu = e^{\int \phi(v)dv}$$

We will see below how to use this equation in various DEs. However, be careful that we need to guess the form of v before hand.

2 Exercises

Problem 1.

$$y' = \frac{1}{e^{-y} - x}$$

Solution. Notice here the differential equation is not homogeneous nor separable. Also, the DE is not exact is difficult to be made exact since $M = 0$. However, we can easily flip the DE turning it into a linear DE.

$$\frac{dx}{dy} + x = e^{-y}$$

Thus, we read off the coefficients, $p(y) = 1$ and $q(y) = e^{-y}$. Notice here that just swapped x and y in the linear DE form. Thus, the general solution can be written as,

$$x(y) = ce^{-\int^{y} p(t)dt} + \int^{y} q(t) e^{-\int_{t}^{y} p(s)ds} dt$$

Thus, $-\int p(t)dt = -t$. We produces,

$$x(y) = ce^{-y} + \int^{y} e^{-t} e^{-(y-t)} dt = ce^{-y} + ye^{-y}$$

Thus, the general solution is,

$$\boxed{x(y) = e^{-y}(c + y)}$$

Problem 2.

$$xy' - 4y + 2x^2 + 4 = 0$$

Solution. Putting this DE into standard form, we can see that its clearly a linear DE,

$$y' \underbrace{-\frac{4}{x}}_{p(x)} y = \underbrace{-2x - \frac{4}{x}}_{q(x)}$$

Next, we find $-\int p(s)ds = 4\ln(s)$. Thus, we can find the general solution as,

$$\begin{aligned} y(x) &= ce^{\ln(x^4)} + \int^x q(t)e^{ln(x^4/t^4)}dt \\ &= cx^4 + \int^x \left(-2t - \frac{4}{t}\right)\frac{x^4}{t^4}dt \\ &= cx^4 + x^2 + 1 \end{aligned}$$

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Thus, the genral solution is,

$$\boxed{y(x) = cx^4 + x^2 + 1}$$

Problem 3.

$$y' = \frac{1}{2}\left(\frac{x + y - 1}{x + 2}\right)^2$$

Solution. This is a differential equation of the form

$$\frac{dy}{dx} = f\left(\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}\right)$$

Thus, this DE is homogeneous given that c_1 and c_2 are zeros. Thus, we have a transformation on x and y to make sure it produces a homogeneous DE. Thus, we transform $x \mapsto u + h$ and $y \mapsto \nu + k$ Thus,

$$\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} \mapsto \frac{a_1u + b_1\nu + (a_1h + b_1k + c_1)}{a_2u + b_2\nu + (a_2h + b_2k + c_2)}$$

Thus, if $(a_2h + b_2k + c_2) = 0$ and $+(a_1h + b_1k + c_1) = 0$ the DE would be homogeneous. In our case we need to solve the system,

$$h + k - 1 = 0$$

$$h + 2 = 0$$

Thus, its clear that $h = -2$ and $k = 3$. Thus, the DE transforms into,

$$\frac{d\nu}{du} = \frac{1}{2}\left(\frac{u + \nu}{u}\right)$$

This is homogeneous of degree 0. Thus, we make the substitution $\nu = uz$. Thus, the DE tranforms into,

$$u\frac{dz}{du} + z = \frac{1}{2}(1 + z) \implies u\frac{dz}{du} = \frac{1}{2}(1 - z)$$

which gives,

$$\int \frac{dz}{1-z} = \frac{1}{2}\int \frac{du}{u} + c \implies -2\ln(1-z) = \ln(u) + c$$

Thus, we would get,

$$(1-z)^{-2} = cu$$

giving the genral solution,

$$\boxed{\frac{1}{1-\frac{y-3}{x+2}} = c(x+2)}$$

Problem 4.

$$xy' + y = x^2y^2$$

Solution. Putting this DE into standard form,

$$y' + \frac{1}{x}y' = xy^2$$

We can see that this is a Bernoulli DE. Thus, one way to solve this DE is by transforming it into a linear DE using substitution $u = y^{1-n}$, where n in this case is 2. Doing so, we end up with the following DE,

$$u' - \frac{1}{x}u = -x$$

Thus, $-\int p(t)dt = \ln(t)$. Thus,

$$y(x) = ce^{\ln(x)} + e^{\ln(x)}\int^{x} (-t)\frac{dt}{t}$$

Thus, the general solutions would be,

$$\boxed{y(x) = x(c-x)}$$

Problem 5.

$$y + 2xe^y + x(1+xe^y)y' = 0$$

Solution. As you can see this problem cannot be homogeneous nor linear. Thus, we check exactness,

$$M_y = \frac{d}{dy}(y + 2xe^y) = 2xe^y + 1$$

$$N_x = \frac{d}{dx}(x(1+xe^y)) = 1 + xe^y + xe^y = 2xe^y + 1$$

Thus, this DE is exact. Since $u_x = M$ Integrating M we get,

$$u(x,y) = xy + x^2e^y + g(y)$$

$u_y = x + x^2e^y + g'(y)$. Thus,

$$x + x^2e^y + g'(y) = x + x^2e^y) \implies g'(y) = 0 \implies g(y) = contant$$

Thus, the general solution is,

$$\boxed{u(x,y) = xy + x^2e^y = c}$$

Problem 6.

$$x^2y + y + 1 + x(1 + x^2)y' = 0$$

Solution. This DE is not homogeneous. Its, however, can be put into a linear form. Nevertheless, its much easier to solve as a non exact problem. Notice that,

$$M_y = x^2 + 1, \qquad N_x = 1 + 3x^2$$

$M_y \neq N_x$, however, to turn this DE into a exact problem with an integrating factor we use equation 2. However, we have first to guess the form of v. Our first guess would be $v = x$. Thus, it must be only a function of x, if not we have to try another guess,

$$\frac{1}{\mu}\frac{d\mu}{dv} = \frac{M_y - N_x}{N} = \frac{-2x^2}{x(1+x^2)} = \phi(x)$$

The function turns out to be a function of x only. Thus, we can easily find the integrating factors,

$$\mu = e^{\int \frac{-2x^2}{x(1+x^2)}dx} = (1 + x^2)^{-1}$$

Thus, the DE turns into,

$$(1 + x^2)^{-1}(x^2y + y + 1) + xy' = 0$$

Since $u_y = N$, we can integrate N to find,

$$u(x, y) = xy + g(x)$$

Thus,

$$u_x = y + g'(x) = \frac{x^2y + y + 1}{1 + x^2} \implies g'(x) = \frac{1}{1 + x^2} \implies g(x) = \tan^{-1}(x)$$

Thus the general solution would be,

$$\boxed{u(x, y) = xy + \tan^{-1}(x) = c}$$

Problem 7.

$$y + x(1 - 3x^2y^2)y' = 0$$

Solution. Clearly, this DE is neither homogeneous, linear or separable. This we check exactness,

$$M_y = 1 \neq N_x = 1 - 9x^2y^2$$

Thus, we are left to find an integrating factor μ. The procedure is to try different guesses of v until one solve the problem. In the following table, we follow this procedure,

v	x	y	xy
$\frac{1}{\mu}\frac{d\mu}{dv}$	$\frac{M_y - N_x}{N}$	$-\frac{M_y - N_x}{M}$	$\frac{M_y - N_x}{yN - xM}$
	$\frac{9x^2y^2}{x(1-3x^2y^2)} \neq \phi(x)$	$-\frac{9x^2y^2}{y} \neq \phi(y)$	$\frac{9x^2y^2}{-3x^3y^3} = -\frac{3}{xy} = \phi(xy)$

Thus, we can find an integrating factor as a function of $v = xy$,

$$\mu = e^{-\int(\frac{3}{v})dv} = e^{-3\ln(v)} = (xy)^{-3}$$

Thus, the DE becomes,

$$\frac{1}{x^3y^2} + \left(\frac{1}{x^2y^3} - \frac{3}{y}\right)y' = 0$$

Since $u_x = M$, $u(x,y) = -\frac{1}{2x^2y^2} + g(y)$. Thus,

$$g'(y) = -\frac{1}{x^2y^3} + \frac{1}{x^2y^3} - \frac{3}{y} \implies g(y) = -3\ln(y)$$

Thus, the general solution is,

$$\boxed{u(x,y) = -\frac{1}{2x^2y^2} - 3\ln(y) = c}$$

Problem 8.

$$(\sin\theta + 2r\cos^2\theta)dr + (-2r^2\sin\theta\cos\theta + r\cos\theta)d\theta = 0$$

Solution. let

$$M = \sin\theta + 2r\cos^2\theta, \qquad N = -2r^2\sin\theta\cos\theta + r\cos\theta$$

$$M_\theta = \cos\theta - 4r\cos\theta\sin\theta$$

$$N_r = \cos\theta - 4r\cos\theta\sin\theta$$

$$F(r,\theta) = \int sin\theta + 2r\cos^2\theta dr = r\sin\theta + r^2\cos^2\theta + g(\theta)$$

$$F_\theta = r\cos\theta - 2r^2\cos\theta sin\theta + g'(\theta) = -2r^2\sin\theta\cos\theta + r\cos\theta$$

$$+g'(\theta) = 0 \to g(\theta) = c$$

$$\boxed{F(r,\theta) = r\sin\theta + r^2\cos^2\theta = c}$$

Problem 9.

$$4x^2yy' = 3x(3y^2+2) + 2(3y^2+2)^2$$

Solution. let $u = 3y^2 + 2 \to u' = 6yy'$ by substitution

$$4x^2 * \frac{1}{6}u' = 6xu + 2u^2$$

$$\frac{2}{3}x^2u' - 3xu = 2u^2$$

$$u' - \frac{9}{4}\frac{1}{x}u = \frac{3}{x^2}u^2 \qquad Bernoulli$$

$$u^{-2}u' - \frac{9}{4x}u = \frac{3}{x^2}$$

let $w = u^{-1} \to w' = -u^{-2}u'$ by substitution

$$-w' - \frac{9}{4x}w = \frac{3}{x^2}$$

$$w' + \frac{9}{4x}w = \frac{-3}{x^2} \qquad Linear$$

$$\mu = \exp \int \frac{9}{4x}dx = e^{\frac{9}{4}\ln(x)} = x^{\frac{9}{4}}$$

$$w = \frac{1}{x^{\frac{9}{4}}}(\int \frac{-3}{x^{-2}}x^{\frac{9}{4}} + c)$$

$$w = \frac{1}{x^{\frac{9}{4}}}(\int -3x^{\frac{1}{4}} + c)$$

$$w = \frac{1}{x^{\frac{9}{4}}}(-3 * \frac{4}{5}x^{\frac{5}{4}} + c)$$

$$u^{-1} = \frac{1}{x^{\frac{9}{4}}}(-3 * \frac{4}{5}x^{\frac{5}{4}} + c)$$

$$\boxed{(3y^2 + 2)^{-1} = \frac{1}{x^{\frac{9}{4}}}(-3 * \frac{4}{5}x^{\frac{5}{4}} + c)}$$

Problem 10.

$$(6x - 3y + 2)dx - (2x - y - 1)dy = 0$$

Solution.

$$\frac{dy}{dx} = \frac{6x - 3y + 2}{2x - y - 1}$$

$$\frac{dy}{dx} = \frac{3(2x - y) + 2}{2x - y - 1}$$

let

$$z = 2x - y \to z' = 2 - y' \qquad substitution$$

$$2 - z' = \frac{3z + 2}{z - 1}$$

$$-z' = \frac{3z + 2}{z - 1} - 2 = \frac{3z + 2 - 2z + 2}{z - 1} = \frac{z + 4}{z - 1}$$

$$\int \frac{1 - z}{z + 4}dz = \int dx$$

$$\int \frac{1}{z + 4} - \frac{z}{z + 4} = x + c$$

$$\int \frac{1}{z + 4} - \frac{z + 4 - 4}{z + 4} = x + c$$

$$\int \frac{1}{z + 4} - 1 + \frac{4}{z + 4} = x + c$$

$$\ln(z + 4) - z + 4\ln(z + 4) = x + c$$

$$\boxed{5\ln(2x - y + 4) - (2x - y) = x + c}$$

Problem 11.

$$(x^2y^2 + y\exp x)dx - \exp x dy = 0$$

Solution.

$$e^x \frac{dy}{dx} - e^x y = x^2 y^2 \qquad Bernolli * y^{-2}$$

$$e^x y^{-2} \frac{dy}{dx} - e^x y^{-1} = x^2$$

let

$$u = y^{-1} \rightarrow u' = -1y^{-2}y' \qquad substitution$$

$$-e^x u' - e^x u = x^2$$

$$u' + u = -x^2 e^{-x} Linear$$

$$\mu = e^{\int dx} = e^x$$

$$u = \frac{1}{e^x}(\int -x^2 e^{-x} e^x + c)$$

$$u = e^{-x}(\int -x^2 + c)$$

$$\boxed{y^{-1} = e^{-x}(-\frac{1}{3}x^3 + c)}$$